

Examination Production & Logistics (Master) Summer Semester 2023 – Heilbronn Campus

July 25, 2023, Moodle On-Site Examination

Please note that the language in this exam is not indicative of answers and (potential) grammar errors do not render an otherwise correct true statement into a false one.

Exercise 1: (10 Points) Multiple Choice Questions

In this exercise, each question answered correctly yields one point. Only one of the answers is correct and there are no negative points for wrongly answered questions.

1.1 Calculate the optimal service level of HP following the assumptions below and those of the case discussed in the class. (1 point)

- Annual holding costs per year are 48% of the value of a printer.
 - Per-unit margin is 22.4%, and 50% of per-unit margin is lost per unit backlogged.
 - Assume one year has 52 weeks.
- a) 90.04 %
 - b) 95.73 %
 - c) 96.04 %
 - d) 92.39 %

1.2 Consider the HP case taught in this module. Suppose that HP DeskJet printers are at the maturity phase of the product life-cycle model. Which of the following is the LEAST appropriate? (1 point)

- a) Customer expectations become lower due to the commoditization.
- b) Superior processes are becoming key success factors.
- c) No more ramp-up problems exist.
- d) Margins become lower due to competition.

1.3 Which of the following is TRUE about the bullwhip effect? (1 point)

- a) It can increase the costs associated with inventories in a supply chain.
- b) It decreases order fluctuations downstream in a supply chain.
- c) It only occurs in beer and semiconductor supply chains.
- d) The bullwhip effect was first noted in the 1970s, is illustrated by the beer game, but thanks to digital transformation is virtually absent today. It is still taught for illustrative purposes.

- 1.4** Suppose supplier S offers payment terms of 60 days to Buyer B and incurs financing costs of 5,750€ when incurring a 10% interest rate. **How large is B's annual spend to S, assuming 360 days per year?** (1 point)
- a) € 517,500
 - b) € 345,000
 - c) € 57,500
 - d) Not enough information to calculate.
- 1.5** Which of the following is MOST suitable for a lean transformation? (1 point)
- a) Produce in larger batches to avoid setup waste (the primary driver of Banpapu Muda).
 - b) Fool-proofing to reduce errors in a process execution.
 - c) Allowing backorders in the inventory models to avoid myopia waste.
 - d) Building large inventory to fulfill even sudden increases in demand (reduce customer-frustration waste).
- 1.6** Your per-order setup cost has decreased by 19%. **What effect will it have on your optimal order quantity, considering all other parameters remain the same?** (1 point)
- a) Optimal order quantity decreases by 44%
 - b) Optimal order quantity decreases by 10%
 - c) Optimal order quantity increases by 44%
 - d) Optimal order quantity increases by 10%
- 1.7** Which of the following concepts BEST describes the case of a firm changing its system from make-to-stock to assemble-to-order? (1 point)
- a) Poka-yoke
 - b) Kaikaku
 - c) Muda
 - d) Kaizen
- 1.8** Which of the following is LEAST LIKELY a cause of the bullwhip effect? (1 point)
- a) Inflated orders
 - b) Everyday low pricing
 - c) Long lead time
 - d) Seasonal discounts
- 1.9** Which of the following describes supply chain finance? (1 point)
- (A) Supply chain finance is often but not exclusively concerned about cash flows along supply chains.

(B) Supply chain finance can enable firms to carry out better operations-related policies.

- a) Both A and B are correct.
- b) A is correct, B is false.
- c) A is false, B is correct.
- d) Both A and B are false.

1.10 Consider statements (A) and (B) about the two extreme cases of the Economic Production Quantity model with its optimal quantity. **Are they correct? (1 point)**

(A) Everything else being equal, the closer the production rate is to infinity, the higher the total cost.

(B) Everything else being equal, the closer the production rate is to the demand rate while maintaining a production rate above the demand rate, the lower the total cost.

- a) Both A and B are correct
- b) A is correct, B is false.
- c) A is false, B is correct.
- d) Both A and B are false.

Exercise 2: Forecasting (10 points)

Consider the following time series to answer the following questions.

Period	Actual Demand	Exponential Smoothing	Moving Average	Linear Regression	Judgemental
Sunday	157				177
Monday	227				252
Tuesday	180				165
Wednesday	260				225
Thursday	419				499
Friday	249				224
Saturday	471				551

- 2.1 Assume $\alpha=0.8$ for exponential smoothing with an initial forecast the same as the demand for Sunday. **Calculate the exponential forecast for Wednesday.** (2 points)
- 2.2 Use actual demand data from the past two days to calculate the moving average forecast for Saturday (i.e., apply moving average forecasting with $N=2$). (2 points)
- 2.3 Suppose an intercept of 105.43 and a slope of 43.75; what is the linear regression forecast for Thursday (assuming Sunday corresponds to time $t=1$)? Round to two decimal places. (2 points)
- 2.4 Calculate the MAD of the entire week of the judgemental forecast. (2 points)
- 2.5 Calculate the MSE of the entire week of the judgemental forecast. Round to two decimal places. (2 points)

Exercise 3: Inventory Management (10 points)

3.1 In the newsvendor model, we have concepts of underage and overage costs. You know that underage costs can result from the opportunity cost of not producing or ordering a unit that could have been demanded and sold. Overage cost can result from the cost per item that was not sold or when we have leftovers from overproduction or ordering too much.

Imagine that your underage cost is 2,500 USD, and your overage cost is 1,700 USD. You know that daily demand is normally distributed with a mean of 40 and a standard deviation of 7 per day.

3.1.1 Assume you plan for a single day. **How many items should you order to minimize your expected costs, assuming all newsvendor model assumptions apply?** (2 points)

3.1.2 **What is the in-stock probability (in %), assuming you have a service level of 63.86%?** (2 points)

3.1.3 Suppose you make an ordering decision for an entire month. Suppose daily demand is independent. Suppose your marketing department informed you that it targets a service level of AT LEAST 80%. **How many units do you need to order per month (30 days) in integer amounts at least?** (2 points)

3.2 Suppose your demand follows a Poisson demand distribution with $\lambda = 25$. **Which one of the following parameter choices refers to a consistent approximation through a normal distribution?** (1 point)

- a) $\mu = 5, \sigma = 5$
- b) $\mu = 25, \sigma = 5$
- c) $\mu = 25, \sigma = 25$
- d) $\mu = 5, \sigma = 25$

3.3 **Which of the following is the most appropriate statement?** (1 point)

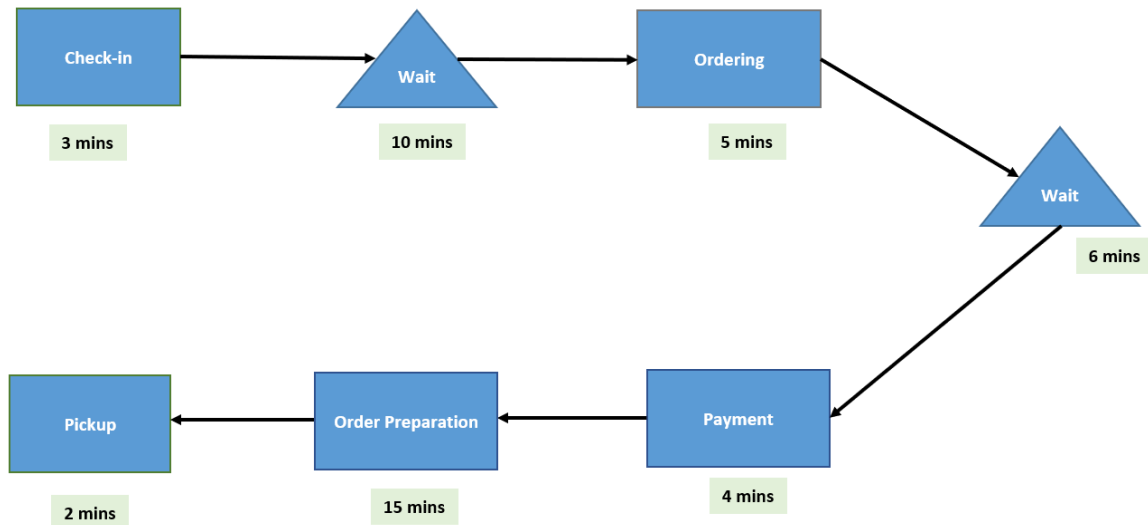
- a) Under the assumption of normally distributed demand, each positive realization of demand has the same probability.
- b) The Poisson distribution is always more suitable than the normal distribution for modelling demand; however, since it is difficult to calculate with the Poisson distribution, we can also assume a uniform distribution without loss of generality.
- c) Irrespective of the demand distribution, the firm should always order the service level. That is, if the service level is 95% and there is a maximum demand of 95, it should order 95 or more units.

- d) A normally distributed random variable can be negative. Demand is always (weakly) positive. Therefore, firms never approximate demand to be normally distributed.

3.4 The demand for a t-shirt is represented as a uniform continuous function of quantities between 3500 and 6000. If the cost per T-shirt = 6 €, the price per T-shirt = 8 €, and unsold units have zero value, **how many units of T-shirts should be ordered?** In other words, what is Q^* ? (2 points)

Exercise 4: Capacity Management (10 points)

The following image shows the flow of customers at a restaurant. The time required for each process is shown.



- 4.1 Suppose there is one worker performing each task (i.e., one who does the check-in, one who takes the orders, one in charge of the payment, ...). **Which of the following is the bottleneck activity?** (1 point)
- a) ordering
 - b) check-in
 - c) order preparation
 - d) waiting

- 4.2 What is the capacity of the restaurant (in customers per hour)? (2 points)

- 4.3 When two employees are involved in the payment activity instead of one, but everything else remains, the capacity of the restaurant will _____. (1 point)
- a) increase by 50%
 - b) remain the same
 - c) increase by 200%
 - d) decrease by 50%

- 4.4 The manager of the restaurant wants to support the order preparation activity but takes 50% more than the trained worker; that is, the manager will prepare orders as well. **Calculate the new capacity of the restaurant** (in customers per hour). (2 points)

- 4.5 Suppose the owner added several workers such that the capacity of the restaurant increased to 11 customers/hour and that it operates at full capacity. **Calculate how many customers are in a steady state on average in the restaurant** (*assuming all activities take place inside the restaurant*) **at any given time.** Round to two decimal places. (2 points)
- 4.6 Suppose the capacity of the restaurant is 10 customers/hour. **Calculate the overall capacity utilization** (in %) **if 6 customers arrive per hour at the restaurant in a steady state.** (2 points)

Exercise 5: Facility Location Problem (10 points)

5.1 Hync GmbH uses aluminum foil rolls to produce four products: chocolate wraps, plates, food containers, and utility cups. Suppose the firm has an aluminum foil roll of 1-meter length. Producing every product unit consumes a certain length of the aluminum foil roll and returns a certain value to Hync GmbH. The table below provides detailed information for all four products.

Product	Length/unit	Value/unit
Chocolate wrap	0.25	1
Plate	0.5	1.5
Food container	0.625	2
Utility cup	0.10	0.5

5.1.1 Which of the following well-known problems best captures Hync GmbH problem to maximize the value from its 1-meter aluminum foil roll? (1 point)

- a) Traveling Salesman Problem
- b) Foil-Rolls Problem
- c) Two-dimension bin-packing Problem
- d) Knapsack Problem

5.1.2 In the same problem, suppose V_i denotes the value/unit of product i , W_i denotes the length of the roll required to produce a unit of product i , and X_i is a binary variable indicating whether Hync produces the product i such that $X_i = 1$ denotes the selection of the product

Formulate the objective function of the problem using the drag-and-drop option. (1 point)

$$\boxed{} \sum_{i=1}^n \boxed{}$$

$$s.t \sum_{i=1}^n \boxed{} \boxed{}$$

$V_i \cdot X_i$
 $W_i \cdot X_i$
Max
Min
< 1
> 1
= 1
 ≤ 1
 ≥ 1

5.1.3 Find the maximum possible value that Hync GmbH could obtain from its 1-meter aluminum foil roll by allowing the firm to produce multiple units of the same product. (2 points)

5.2 Consider the traveling salesmen problem formulation below. Assign the most appropriate terms to each row; i.e., put the most suitable label to the objective function and each constraint. (2 points)

$$\min \sum_{(i,j) \in E} c_{ij} x_{ij}$$

Drag answer here

$$\sum_{i \in V: (i,j) \in E} x_{ij} + \sum_{i \in V: (j,i) \in E} x_{ji} = 2, \quad j \in V$$

Drag answer here

$$\sum_{(i,j) \in E: i \in S, j \notin S} x_{ij} + \sum_{(j,i) \in E: i \in S, j \notin S} x_{ji} \geq 2, \quad S \subset V, 2 \leq |S| \leq \left\lfloor \frac{|V|}{2} \right\rfloor$$

Drag answer here

- a) Conservation of flow constraint.
- b) Minimizing the cost of traversing vertices.
- c) Minimizing the cost of traversing edges.
- d) Minimization of shadow costs.
- e) Avoiding double use of edges constraint.
- f) Subtour elimination constraint.
- g) Subtour avoidance constraint.
- h) City visiting maximization.

5.3 Consider a setting where you have four plants, namely P1, P2, P3, and P4, and you must serve five customers C1, C2, C3, C4, and C5. The cost to open P1, P2, P3, and P4 are 5, 0, 3, and 4 units, respectively. P1 can serve all customers, P2 can serve C1, C2 and C4, P3 can serve C4 and C5, and P4 can serve C3 and C5. All customers are to be served.

5.3.1 Which option below corresponds to the solution using the set covering heuristic discussed in this course? (2 points)

- a) open exactly plant P1, the only plant that serves all customers, leading to total costs of 5.
- b) open exactly P2 and compensate customers C3, C4, and C5 as long as the compensation costs less than opening plant P1, leading to total costs likely below 4.
- c) open exactly plants P2 and P4, leading to total costs of 4, which is also the optimal solution.
- d) open all plants, leading to total costs of 3 = min(5,3,4).

5.3.2 What is the minimum total cost that you incur in the heuristics solution? (2 points)

Exercise 6: Kristen's Cookie Company (5 points)

6.1 Suppose the current bottleneck of Kristen Cookies was the oven with a capacity of 6 dozen per hour. Suppose the mixing process takes 6 minutes, and it can mix ingredients for one, two, or three trays simultaneously. But it can only mix ingredients for multiple trays if they all follow the same recipe for the same dough. So, if a customer places an order for a special dough for two dozen of cookies, for instance, the mixer could mix both simultaneously; but if there are orders for different doughs, those need to be mixed sequentially, requiring 6 minutes each. Some customers place orders for more than one dough, and whether this happens or not is random. Now suppose Kristen Cookies obtains a second oven that runs in parallel with the same capacity as the first one; **which of the following is the most appropriate conclusion?** (1 point)

- a) There is no benefit in having the second oven.
- b) It would be valuable to have a buffer/inventory of readily mixed and/or spooned cookies.
- c) There is no benefit in having the second oven.
- d) With the second oven, the capacity doubles.

6.2 The cookies are prepared once the order is received at Kristen Cookies. The following table shows the sequence of activities that must be followed to fulfill the order, the time required for each activity involved in preparing one dozen cookies, and the resource required for each activity.

	Activity	Time Required (mins)	Resource Required
1.	Confirming order and printing order details	1	Worker
2.	Mixing ingredients	6	Mixer
3.	Spooning	2	Worker
4.	Baking	10	Oven
5.	Cooling	5	-
6.	Packing, Payment, and Delivery	3	Worker

- You are the sole worker and can perform one activity at a time.
- You have more than one mixing bowl for the mixer and more than one tray for the oven.
- The mixer can mix ingredients for a maximum of 12 cookies of the same type.
- The oven can bake a maximum of 12 cookies.

Calculate the minimum time required (in mins) to complete two different orders of one dozen cookies each that are received at the same

time. Assume that no orders from the past or present are pending and neglect the transition time from one activity to the next. (2 points)

6.3 Kristen Cookies Company is using a _____ production system for fulfilling its customer's orders. The bottleneck for the company is the _____. Suppose Kristen Cookies becomes increasingly popular and seeks to sell more cookies to meet the increasing demand, then it should primarily consider _____. Even if sometimes tempting, it would be risky to switch to a _____ production system; doing so could threaten the business model as their cookies would no longer be _____. (2 points)

- a) make-to-stock
- b) tasty
- c) mixing process
- d) lack of human resources
- e) buying a larger oven
- f) baking process
- g) engineer-to-order
- h) accelerating the payment process
- i) inexpensive
- j) training the current workforce
- k) hiring new employees
- l) make-to-order
- m) customized

Exercise 7: OceanCove (5 points)

7.1 Suppose there is a restaurant called Cocean Ove which resembles the famous one called OceanCove, as discussed in class. The restaurants are quite similar, except for some differences: Cocean Ove has the following capacities:

Activity	Capacity (meals/hour)
Fish-Grilling	100
Fish-Frying	200
French Fry-Frying	400
Expediting	plenty
Assembling	180
Order-taking, Delivery, and Billing	120
Lunchtime Dining Area	270

And, at Cocean Ove, people order fried fish as often as they order grilled fish. Still, every customer orders exactly one meal. **Calculate the overall capacity.** (2 points)

7.2 The waiters at Cocean Ove use paper slips to write down customers' orders, who then take the slips to the kitchen and give them to the chefs. The chef prepares the food accordingly, one order at a time. **Passing on the order information through paper slip can best be related to the lean concept** _____. (1 point)

- a) Kaizen
- b) Keiretsu
- c) Kaikaku
- d) Kanban

7.3 Suppose the average customer arrival rate and the average order preparation rate is 2 orders/minute at Cocean Ove. Assume that each customer orders exactly once. **Calculate the average waiting time (in minutes) for a customer to receive their order if there are 38 orders waiting simultaneously.** (2 points)